Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT		
S. No.	**University ID**	**Name**
1	2420090138	UTAKRSH PANKAJ
2	2420090083	SARU DHANUSH
3	2420030778	RITHISH RAO
4		
Name of the Guide		

Abstract:

As the number of connected devices and digital services continues to grow, ensuring that only trusted users and devices can access sensitive resources has become an important security challenge. Traditional access control systems mainly depend on usernames, passwords, and predefined permissions. However, these mechanisms alone may not be sufficient when a valid account is accessed from an unknown, compromised, or suspicious device.

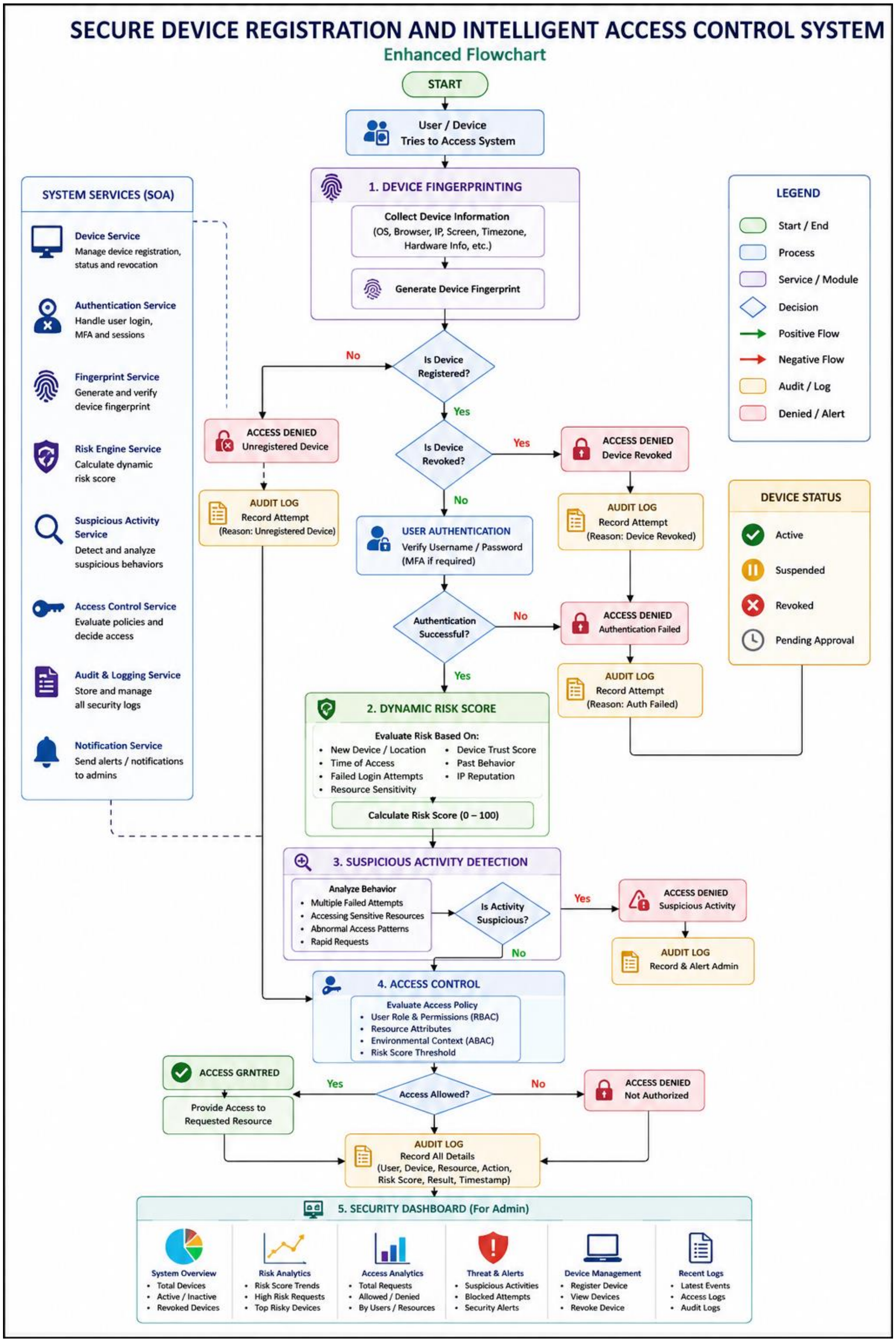
The **Secure Device Registration and Intelligent Access Control System** is proposed to provide an additional layer of security by considering both the **user and the device** during access decisions. The system allows administrators to securely register devices and maintain their status. When a user attempts to access a resource, the system verifies the user's credentials and generates a **device fingerprint** to determine whether the requesting device matches the registered device. Revoked or unrecognized devices are automatically denied access.

The system also introduces a **dynamic risk scoring mechanism** that evaluates factors such as device trust, failed login attempts, access patterns, resource sensitivity, and other contextual information. Based on the calculated risk, the system can allow or deny access and identify

potentially suspicious activity. Administrators can revoke compromised devices and monitor security events through a centralized **security dashboard** containing device information, access attempts, risk levels, alerts, and audit logs.

The system follows a **Service-Oriented Architecture (SOA)**, where functions such as device registration, authentication, risk evaluation, access control, suspicious activity detection, and auditing are implemented as independent services that communicate through APIs. The proposed system aims to provide a flexible, scalable, and security-focused approach to managing device trust and controlling access to protected resources.

Project Architecture/Project Overview:


SECURE DEVICE REGISTRATION AND INTELLIGENT ACCESS CONTROL SYSTEM
Enhanced Flowchart
START
User / Device
Tries to Access System
1. DEVICE FINGERPRINTING
Collect Device Information
(OS, Browser, IP, Screen, Timezone, Hardware Info, etc.)
Generate Device Fingerprint
Is Device Registered?
No
Yes
ACCESS DENIED
Unregistered Device
AUDIT LOG
Record Attempt
(Reason: Unregistered Device)
Is Device Revoked?
Yes
No
ACCESS DENIED
Device Revoked
AUDIT LOG
Record Attempt
(Reason: Device Revoked)
USER AUTHENTICATION
Verify Username / Password
(MFA if required)
Authentication Successful?
No
Yes
ACCESS DENIED
Authentication Failed
AUDIT LOG
Record Attempt
(Reason: Auth Failed)
SYSTEM SERVICES (SOA)
Device Service
Manage device registration, status and revocation
Authentication Service
Handle user login, MFA and sessions
Fingerprint Service
Generate and verify device fingerprint
Risk Engine Service
Calculate dynamic risk score
Suspicious Activity Service
Detect and analyze suspicious behaviors
Access Control Service
Evaluate policies and decide access
Audit & Logging Service
Store and manage all security logs
Notification Service
Send alerts / notifications to admins
LEGEND
Start / End
Process
Service / Module
Decision
Positive Flow
Negative Flow
Audit / Log
Denied / Alert
DEVICE STATUS
Active
Suspended
Revoked
Pending Approval
2. DYNAMIC RISK SCORE
Evaluate Risk Based On:
• New Device / Location
• Time of Access
• Failed Login Attempts
• Resource Sensitivity
• Device Trust Score
• Past Behavior
• IP Reputation
Calculate Risk Score (0 – 100)
3. SUSPICIOUS ACTIVITY DETECTION
Analyze Behavior
• Multiple Failed Attempts
• Accessing Sensitive Resources
• Abnormal Access Patterns
• Rapid Requests
Is Activity Suspicious?
Yes
No
ACCESS DENIED
Suspicious Activity
AUDIT LOG
Record & Alert Admin
4. ACCESS CONTROL
Evaluate Access Policy
• User Role & Permissions (RBAC)
• Resource Attributes
• Environmental Context (ABAC)
• Risk Score Threshold
ACCESS GRNTRED
Provide Access to Requested Resource
Access Allowed?
Yes
No
ACCESS DENIED
Not Authorized
AUDIT LOG
Record All Details
(User, Device, Resource, Action, Risk Score, Result, Timestamp)
5. SECURITY DASHBOARD (For Admin)
System Overview
• Total Devices
• Active / Inactive
• Revoked Devices
Risk Analytics
• Risk Score Trends
• High Risk Requests
• Top Risky Devices
Access Analytics
• Total Requests
• Allowed / Denied
• By Users / Resources
Threat & Alerts
• Suspicious Activities
• Blocked Attempts
• Security Alerts
Device Management
• Register Device
• View Devices
• Revoke Device
Recent Logs
• Latest Events
• Access Logs
• Audit Logs